

Physiology and Mathematics of Inhalational Anesthesia

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Abstract

Inhalational anesthetics reach the brain through a coupled respiratory and circulatory system. Their speed of onset and recovery is determined not merely by a vaporizer setting, but by fresh-gas delivery, rebreathing, alveolar ventilation, blood solubility, cardiac output, and uptake by tissues. This document develops a teaching model with a breathing circuit, alveoli, and vessel-rich, muscle, and fat tissue groups. It documents the assumptions used by the accompanying interactive simulator and explains why it is not a clinical gas-delivery calculator.

1 Partial pressure, fraction, and potency

For an inert volatile anesthetic, pharmacologic effect is governed most directly by partial pressure (or tension), not by total anesthetic mass. At a fixed barometric pressure, gas fraction F and partial pressure P are proportional: $P = FP_B$. Thus fractions can be used as normalized partial pressures in a teaching model.

Minimum alveolar concentration (MAC) is a population reference for anesthetic potency: conventionally, the alveolar concentration preventing movement to a standard noxious stimulus in 50% of subjects. It is not a universal measure of adequate anesthesia in an individual, and it varies with age, temperature, concomitant drugs, and clinical context. A displayed MAC fraction in the simulator is therefore only F_A/MAC_{ref} .

2 The path from vaporizer to tissues

The model is deliberately physiologic rather than an IV-style mammillary model:

vaporizer $\rightarrow$ breathing circuit $\rightarrow$ alveoli $\rightarrow$ arterial blood $\rightarrow$ {vessel-rich, muscle, fat} $\rightarrow$ venous blood. (1)

The vessel-rich group is a composite teaching compartment representing highly perfused tissues, including the brain. It is not a measured brain concentration. Muscle and fat receive less blood flow but can become important reservoirs during long exposures.

3 Circuit and alveolar mass balances

Let F_V , F_I , and F_A be the vaporizer, circuit-inspired, and alveolar fractions. Let V_C be circuit volume, V_A functional residual capacity, $\dot{V}_{FGF}$ fresh-gas flow, and $\dot{V}_A$ alveolar ventilation. A well-mixed circuit gives

$$V_C \frac{dF_I}{dt} = \dot{V}_{FGF}(F_V - F_I) - \dot{V}_A(F_I - F_A). \quad (2)$$

The first term replaces circuit gas with fresh gas; the second transfers gas between circuit and alveoli. The alveolar balance is

$$V_A \frac{dF_A}{dt} = \dot{V}_A(F_I - F_A) - U_{pulm}, \quad (3)$$

where pulmonary uptake is approximated by

$$U_{pulm} = \dot{Q}\lambda_{b:g}(F_A - F_{\bar{v}}). \quad (4)$$

Here $\dot{Q}$ is cardiac output, $\lambda_{b:g}$ is the blood:gas partition coefficient, and $F_{\bar{v}}$ is mixed venous anesthetic partial-pressure fraction. Equation 4 summarizes the classic determinants of uptake: a greater alveolar–venous gradient, higher cardiac output, or greater blood solubility removes more anesthetic from alveoli and slows the rise of F_A/F_I .

4 Tissue groups and venous return

For a tissue group j with volume V_j , perfusion $\dot{Q}_j$, tissue:blood partition coefficient $\lambda_{j:b}$, and normalized tissue tension F_j , a perfusion-limited approximation is

$$\frac{dF_j}{dt} = \frac{\dot{Q}_j}{V_j\lambda_{j:b}}(F_A - F_j). \quad (5)$$

The model uses three groups: vessel-rich, muscle, and fat, with $\sum_j \dot{Q}_j = \dot{Q}$. Mixed venous fraction is perfusion weighted,

$$F_{\bar{v}} = \frac{1}{\dot{Q}} \sum_j \dot{Q}_j F_j. \quad (6)$$

The coupled system in Eqs. 2–6 produces familiar behavior. Early in an anesthetic, venous tension is low and uptake is high. Later, vessel-rich tissue equilibrates, decreasing uptake; muscle and fat can continue to accumulate anesthetic and contribute to delayed washout.

5 Wash-in, washout, and solubility

Low $\lambda_{b:g}$ means that little anesthetic must dissolve in blood before alveolar partial pressure rises. This produces a more rapid rise in F_A/F_I and, after vaporizer discontinuation, more rapid elimination. Higher ventilation accelerates circuit/alveolar replacement, whereas higher cardiac output increases uptake from alveoli and can slow alveolar wash-in. Fresh-gas flow is particularly important when circuit volume and rebreathing are represented: low flow delays the approach of F_I to the vaporizer fraction.

Vaporizer-off washout is not simply exponential with one time constant. Circuit replacement, pulmonary elimination, redistribution from vessel-rich tissue, muscle, and fat, and their different perfusion/solubility time constants all contribute. This is why a teaching model should display both end-tidal fraction and tissue-equivalent states.

6 Worked teaching example: sevoflurane wash-in and washout

Consider an illustrative, not patient-specific, controlled-ventilation scenario. Select sevoflurane in the simulator and use the inputs in Table 1. The values describe the simplified apparatus-and-patient model; they are not a prescription, a recommended vaporizer setting, or a substitute for monitored clinical care. In particular, clinical inspired and end-tidal concentrations must be selected and adjusted by qualified clinicians using the actual machine, patient, and monitoring context.

At the start, F_I , F_A , and all tissue-equivalent fractions are zero. The 2.0% vaporizer setting first fills the circuit, so the inspired fraction rises before the alveolar fraction. In the current simulator implementation, the example produces approximately $F_A = 1.08\%$ at 5 minutes, 1.39% at 15 minutes, and 1.56% just before the 60-minute vaporizer-off time. The latter corresponds to about 0.74 of the displayed reference MAC (1.56/2.1). The vessel-rich equivalent rises rapidly behind F_A (about 1.56% at 60

Table 1: Illustrative sevoflurane simulator inputs.

Simulator input	Example value
Agent	Sevoflurane ($\lambda_{b:g} = 0.65$; reference MAC = 2.1%)
Vaporizer fraction	2.0%
Fresh-gas flow	4.0 L/min
Alveolar ventilation	4.0 L/min
Functional residual capacity	2.5 L
Breathing-circuit volume	2.0 L
Cardiac output	5.0 L/min
Vaporizer off time	60 min
Total simulation time	120 min

minutes), whereas muscle and fat rise much more slowly (about 0.62% and 0.05%, respectively). These numerical values are outputs of this teaching model, not predicted clinical end-tidal concentrations.

At 60 minutes the simulator changes F_V from 2.0% to zero. Fresh gas and ventilation then clear the circuit and alveoli: in this example F_A falls to roughly 0.93% one minute later and approaches zero by about 75 minutes. The vessel-rich group follows during this early washout, while muscle and fat remain nonzero because they are slower compartments. The first plot therefore shows the delivery and rapid end-tidal response; the tissue plot makes the delayed reservoir behavior visible; and the uptake plot is highest early, when the alveolar–venous gradient is greatest, then declines as the compartments equilibrate.

Useful counterfactuals are to increase fresh-gas flow or alveolar ventilation, which shortens circuit/alveolar turnover in this model, and to increase cardiac output, which increases early pulmonary uptake and tends to slow the rise of F_A . Changing only the vaporizer-off time demonstrates why a longer exposure loads the slower tissue groups more strongly. These are qualitative physiologic lessons; the exact sizes and time courses should not be extrapolated to clinical management.

7 Numerical implementation

The webpage integrates the five state variables $[F_I, F_A, F_{VRG}, F_{muscle}, F_{fat}]^T$ with fourth-order Runge–Kutta integration at a fixed short time step. Inputs are held constant across a step; after the chosen vaporizer-off time, F_V is set to zero. Fractions are constrained to nonnegative values. The implementation is intended to reveal causal relationships, not to reproduce a particular anesthesia machine or commercial gas analyzer.

8 Important omissions and safety limitations

The simulator omits airway dead space, spontaneous effort, V/Q mismatch, shunt, diffusion limitation, nonlinear tissue partitioning, metabolism, agent interaction, nitrous-oxide concentration and second-gas effects, altitude, humidification, ventilator mechanics, scavenging, vaporizer physics, and monitoring. It also uses generic tissue-group volumes and perfusion fractions rather than a patient-specific validated model. Therefore it must not be used to select vaporizer concentrations, judge anesthetic depth, manage emergence, or guide patient care.

References

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